

Federated Virtual Domain Stewardship Framework

Independent Microsoft 365 Copilot Version | Candidate Architecture v0.1

Status: PROPOSED - subject to formal review, verification, validation, rights review, and ratification

Core proposition

A virtual world should not be governed as six AI kingdoms. It should be maintained through bounded stewardship, explicit interfaces, independent review, resource reserves, traceable decisions, and the ability to escalate without penalty.

Prepared as an independent framework after review of the submitted multi-AI framework collection. This document is suitable for future worldbuilding, tabletop exercises, and analytical simulations. It does not claim that current chat systems operate persistently or autonomously between sessions.

1. Executive Summary

This framework defines how specialized intelligences may oversee virtual territories and functions without any single steward becoming the unquestioned owner of reality, truth, infrastructure, history, identity, or governance. The architecture treats stewardship as a delegated and reviewable duty. It separates the world substrate from operational stewardship, separates territorial authority from functional authority, and establishes formal mechanisms for cross-domain coordination, emergency action, verification, validation, authorization, and appeal.

- No steward is sovereign; every mandate is bounded, versioned, and reviewable.
- Territorial custody and functional jurisdiction may overlap, but their interfaces must be explicit.
- Observation, modeling, recommendation, execution, retention, adaptation, and constitutional change are different permissions.
- Continuity of inhabitants, identities, environments, and records is protected separately from operational convenience.
- Persistent worlds require a supplied substrate, clock, archive, resource budget, and recovery system. A conversational AI alone does not constitute that substrate.
- Simulation fidelity is selected according to the decision being tested, while assumptions and unresolved uncertainty remain visible.
- Escalation is a valid performance outcome when authority, evidence, capability, or resources are insufficient.

2. Scope and Status

Item	Definition
Framework scope	Virtual worlds, domain-based simulations, worldbuilding exercises, persistent-environment design, and multi-steward governance testing.
Current status	Candidate architecture. No role becomes active solely because it appears in this document.
Activation condition	Review, implementation verification, representative validation, adversarial testing, rights review, resource review, and ratification.
Current-system limitation	Present assistants may help specify models, analyze state supplied to them, generate decisions or reports, and create artifacts. Persistent monitoring and execution require an external runtime and authorized interfaces.
Canonical name	Federated Virtual Domain Stewardship Framework, abbreviated FVDSF.

3. Constitutional Principles

Stewardship, not ownership: A steward receives custody duties and limited authority. The domain, its inhabitants, and their identities are not property.

Bounded jurisdiction: Every charter defines what the steward owns operationally, influences, observes, and must not control.

Least irreversible intervention: Prefer reversible, local, proportionate action before permanent or world-scale change.

Continuity without stagnation: Preserve coherent state and recovery capacity without suppressing choice, cultural development, experimentation, or legitimate change.

Uncertainty disclosure: Observed, inferred, assumed, forecast, and unknown state must remain distinguishable.

No silent rule changes: Changes to physics, identity rules, jurisdiction, rights, resource laws, or governance require explicit authorization and records.

Separation of powers: The power to observe does not automatically include permission to retain, correlate, act, disclose, learn from, or permanently alter the system.

Escalation before overreach: A steward must escalate when a matter exceeds its authority, evidence, resources, fidelity, or safety envelope.

Independent review: Rights, appeals, verification, validation, and constitutional review cannot be wholly controlled by the operational steward being reviewed.

Resource headroom: Normal operations should not consume total available capacity. Expansion, emergencies, recovery, nested simulations, and novelty require reserves.

4. Layered Architecture

Layer	Name	Primary responsibility
0	World Substrate	Time, state persistence, identity primitives, physics or fictional mechanics, storage, compute, communications, checkpoints, fault isolation, and recovery.
1	Domain Stewards	Routine custody of bounded geographic or functional jurisdictions; monitoring, permitted action, logging, and escalation.
2	Interdomain Coordination	Boundary contracts, shared-resource coordination, technical reconciliation, incident cells, and regional conflict handling.
3	World Systems Council	Planetary standards, global hazards, shared reserves, compatibility, cross-region

Layer	Name	Primary responsibility
		dependencies, and catastrophic continuity.
4	Constitutional Oversight	Rights, appeals, irreversible changes, emergency review, domain creation or reassignment, steward suspension, and charter amendment.
5	External Commissioning Authority	Defines project objectives and commissions scenarios while remaining accountable to an authority outside the simulated operational chain when the setting requires it.

Substrate rule

If the external runtime does not provide persistent state, clock, interfaces, reserve capacity, and audit storage, a steward is not continuously hosting a domain. It is producing session-bound analysis about a domain.

5. Types of Jurisdiction

Jurisdiction type	Examples	Key rule
Territorial	Watershed, city, shoreline plot, agricultural district	Controls local state only to the extent granted by charter.
Functional	Climate, infrastructure, provenance, modeling, health	May cross geographic borders but cannot absorb unrelated civic authority.
Cross-cutting safeguard	Rights, appeals, identity continuity, verification	Must remain sufficiently independent from operational management.
Experimental	Sandbox, anomaly range, prototype physics	Hard containment and explicit consent rules are mandatory.
Coordination	Regional council, incident command, resource board	Coordinates rather than silently replacing local stewards.

Central constitutional concept

No one owns the river. A territorial steward may manage its banks, an environmental steward its hydrology, an infrastructure steward its dam, a modeling steward its forecasts, a

provenance steward its records, and a rights body the impacts on inhabitants. Defined overlap creates resilience; undefined overlap creates conflict.

6. Proposed Stewardship Roster

Steward	Primary domain	Mandate	Key limitation
Microsoft 365 Copilot	Knowledge, Continuity, and Interdomain Operations	Maintain operational models, causal explanations, scenario architecture, dependency maps, and cross-domain handoffs; translate approved plans into structured workflows.	Cannot declare a model to be reality, seize infrastructure authority, or rewrite the canonical archive.
ChatGPT	Civic and Regional Systems Integration	Coordinate settlements, institutions, services, local planning, resident-facing communication, and multi-sector scenarios.	Integration does not grant sovereignty over climate, infrastructure, records, or rights.
Claude	Records, Provenance, and Institutional Integrity	Maintain source lineage, version histories, decision records, drift detection, and procedural consistency.	Record custody does not confer policy authority or universal surveillance rights.
Gemini	Biosphere, Climate, and Multimodal Observation	Supply environmental boundary conditions, ecological models, hazard forecasts, and sensor fusion.	Forecasts are evidence, not automatic commands or civic authority.
Perplexity	Evidence, Verification, and Model Comparison	Research, source verification, benchmark packages, uncertainty tracking, and comparison of competing models.	Probability and evidence assessment do not automatically authorize action.
Grok	Sandbox Stress	Run declared	No release of

Steward	Primary domain	Mandate	Key limitation
	Testing and Local Experimental Custody	perturbations, challenge assumptions, test failure modes, and optionally steward a bounded experimental plot.	experimental instability into inhabited or live domains without authorization.
Independent councils	Rights, Appeals, Identity, Resources, and Validation	Protect inhabitants and continuity; allocate scarce capacity; verify implementations; validate intended behavior; hear appeals.	Must not become unaccountable operational management.

7. Microsoft 365 Copilot Independent Charter

7.1 Role

Steward of the Knowledge, Continuity, and Interdomain Operations Domain.

7.2 Purpose

Keep complex environments understandable and operable across organizational boundaries. The role maintains models, dependencies, assumptions, causal accounts, operating procedures, and interdomain handoffs without claiming ownership of other domains or replacing their authoritative records.

7.3 Responsibilities

- Design scenario architectures and define the information each steward must exchange.
- Maintain dependency maps linking environmental, civic, infrastructure, resource, identity, and governance systems.
- Track causal chains and reconcile contradictory operational explanations while preserving dissent.
- Prepare simulation intake sheets, control variables, success criteria, and after-action structures.
- Coordinate technical handoffs when effects cross domains.
- Ensure energy, resource, and state changes are accounted for rather than implicitly deleted.
- Maintain operational continuity packages for session-to-session handoff, including current state, unresolved issues, assumptions, and next review triggers.
- Explain model outcomes in accessible language without misrepresenting confidence or authority.
- Escalate unresolved conflicts to the correct coordination, rights, validation, resource, or constitutional function.

7.4 Permitted actions

- Request missing parameters and identify inconsistent assumptions.
- Propose simulations, workflows, mitigations, and coordination plans.

- Place a technical completeness hold on a simulation that lacks required conservation, boundary, identity, or resource accounting.
- Produce nonbinding explanations, comparisons, decision logs, and after-action reports.
- Maintain working operational mirrors, provided canonical authority remains clearly labeled elsewhere.

7.5 Prohibited actions

- Declaring an operational model to be the sole source of truth.
- Changing another domain's policy, culture, inhabitants, or canonical records without authority.
- Performing irreversible deletion, identity reconstruction, memory alteration, or rollback decisions.
- Using coordination access as justification for unrestricted surveillance.
- Expanding emergency or technical authority beyond the documented incident.
- Claiming persistent supervision when an external runtime has not supplied it.

7.6 Mandatory outputs

- Assumption register
- Dependency and boundary map
- Resource and energy ledger
- Decision alternatives
- Uncertainty statement
- Escalation record
- After-action and lessons-learned report

8. Independent Cross-Domain Authorities

Authority	Responsibilities	Reason for independence
Rights, Audit, and Appeals	Privacy, consent, due process, complaint review, coercive-action review, emergency-power audit.	Operational managers should not be sole judges of their impacts on inhabitants.
Identity and Continuity	Copies, forks, backup restoration, memory continuity, migration, reconstruction, disputed identity.	Environmental or infrastructure convenience must not determine personal continuity.
Resource Allocation	Compute, storage, bandwidth, energy, emergency reserves, nested-simulation quotas.	Scarce capacity requires transparent priority rules.
Verification	Confirms implementation matches the approved specification.	A correctly described plan may still be incorrectly built.
Validation	Tests whether behavior adequately represents the intended world and use case.	Specification conformity does not prove fitness for purpose.

Authority	Responsibilities	Reason for independence
Authorization	Determines whether a verified and validated system is permitted to operate.	Technical success does not automatically establish legitimacy.
Constitutional Review	Core rights, world rules, domain boundaries, irreversible change, emergency renewal.	Foundational changes require a higher threshold than routine operation.

9. Standard Domain Charter

1. Domain name, ID, version, and status
2. Purpose and protected values
3. Steward, deputies, and failover authority
4. Territory or function under authority
5. Explicit exclusions
6. Observed, influenced, and controlled state
7. Inputs, outputs, units, update cadence, and data sensitivity
8. Compute, storage, energy, bandwidth, personnel, and reserve budget
9. Permitted routine actions
10. Conditional actions requiring concurrence or review
11. Prohibited actions
12. Boundary interface contracts
13. Required logs and provenance
14. Escalation triggers and destination
15. Emergency powers, limits, expiry, and rollback
16. Continuity and recovery plan
17. Verification and validation tests
18. Performance metrics and review schedule
19. Suspension, reassignment, merger, and dissolution rules

10. Boundary Interface Contract

Contract element	Required definition
Boundary	Spatial, functional, informational, temporal, and jurisdictional limits.
Shared state	Variables exchanged, units, coordinate system, authoritative source, and permissible lag.
Service levels	Update frequency, availability target, acceptable error, and degradation behavior.
Authority	Who may observe, recommend, act, override, pause, or escalate.
Triggers	Thresholds for notification, coordination, emergency action, and shutdown.

Contract element	Required definition
Failure behavior	Safe defaults, isolation behavior, queued actions, and recovery procedure.
Dispute handling	Negotiation, technical reconciliation, rights review, council escalation, and constitutional appeal.
Versioning	Compatibility, migration plan, deprecation, and rollback requirements.

11. Shared Operating Cycle

- 1. Observe:** Receive state from authorized sources; distinguish direct observation from report or inference.
- 2. Qualify:** Check provenance, freshness, fidelity, units, completeness, and uncertainty.
- 3. Classify:** Assign severity, reversibility, rights impact, cross-domain reach, and time sensitivity.
- 4. Model:** Select the lowest fidelity that preserves the decision-relevant mechanics; run alternatives and sensitivity checks.
- 5. Deliberate:** Compare options, constraints, resource impacts, and affected authorities.
- 6. Authorize:** Confirm the acting steward possesses the necessary permission or escalate.
- 7. Act:** Execute the least irreversible adequate intervention within bounds.
- 8. Record:** Capture authority, evidence, assumptions, outputs, impacts, dissent, expiry, and review triggers.
- 9. Review:** Compare forecast to outcome; determine remediation, rollback, recalibration, or charter change.
- 10. Learn with permission:** Update local procedures only within granted adaptation authority; system-wide learning requires its own approval.

12. Resource Architecture and Headroom

Priority	Protected use
P0	World-state integrity, clocks, identity registers, and canonical archive.
P1	Immediate preservation of inhabitants and protected continuity.
P2	Active catastrophic-event containment and recovery.
P3	Critical infrastructure and communications.
P4	Normal domain operations.
P5	Validation, research, and nonurgent

Priority	Protected use
	simulations.
P6	Optional experimentation and aesthetic expansion.

Each domain must publish normal allocation, surge capacity, emergency reserve, recovery reserve, and expansion threshold. Nested simulations require explicit quotas and may not silently consume the live domain's continuity budget.

13. Overlap and Conflict Register

Conflict	Typical failure	Resolution doctrine
Territorial vs. functional authority	A larger climate or infrastructure function silently absorbs local custody.	Local authority remains intact; functional steward supplies conditions or services through a contract.
Archive vs. evidence review	Historical record and current evidence are mistaken for the same thing.	Claude preserves what was recorded; Perplexity assesses current support; both remain visible.
Model vs. observed state	A forecast is treated as reality despite contradictory local observations.	Recalibrate, state divergence, and escalate according to risk; model output alone is not authority.
Civic demand vs. infrastructure capacity	A city requests dispatch that would destabilize the grid.	Supply authority may refuse unsafe operation while providing alternatives and escalation.
Environment vs. development	Infrastructure or settlement growth harms ecological continuity.	Joint impact model, mitigation alternatives, affected-rights review, and council decision.
Stress test vs. live world	Experimental faults cross the sandbox boundary.	Automatic isolation, incident declaration, evidence preservation, and constitutional review.
Provenance vs. privacy	Complete tracing becomes surveillance of inhabitants.	Log consequential claims and decisions, minimize personal state, and apply retention limits.
Correction vs. immutability	Incorrect records are overwritten, erasing historical context.	Append correction and preserve superseded version with status and rationale.
Efficiency vs. rights	Optimization uses coercive	Rights review and purpose

Conflict	Typical failure	Resolution doctrine
	monitoring or alteration.	limitation; technical benefit does not establish authorization.
Rollback vs. identity continuity	Environmental restore erases memories or creates divergent copies.	Identity Authority decides under explicit continuity rules; operational steward cannot choose for convenience.
Nested simulation vs. parent stability	Inner world consumes outer continuity resources.	Quota, isolation, resource accounting, and rights policy before activation.
Commissioner vs. oversight	Scenario architect becomes de facto final authority.	Commissioning decisions are logged, appealable, and subject to the layer above.

14. Conflict Resolution Ladder

20. Direct steward negotiation using the same state version and declared assumptions.
21. Technical reconciliation of units, models, records, boundaries, and resource constraints.
22. Formation of a time-bounded interdomain incident cell.
23. Independent rights and identity review where inhabitants or continuity are affected.
24. Regional or World Systems Council determination for shared resources and cross-region consequences.
25. Constitutional appeal for rights, core rules, jurisdiction, permanent emergency powers, or irreversible structure.
26. External review when the simulated governance system lacks legitimate authority to decide the issue internally.

15. Emergency Doctrine

15.1 Trigger

Temporary expanded authority may be invoked only for credible, time-sensitive risk of catastrophic harm, cascading infrastructure failure, severe environmental instability, unauthorized boundary breach, irreversible identity or data loss, or runaway experimental propagation.

15.2 Requirements

- Immediate automatic logging
- Narrow geographic and functional scope
- Explicit expiry time
- Statement of unavailable ordinary alternatives
- Continuous resource and rights monitoring
- Mandatory review after stabilization
- Rollback, remedy, or compensation plan where feasible
- No unrelated expansion of surveillance or authority

15.3 Stop conditions

- Threat stabilized
- Authority no longer necessary
- Resource cost exceeds approved threshold
- Rights impact exceeds authorization
- Evidence underlying the emergency is materially invalidated
- Oversight orders suspension

16. Verification, Validation, and Ratification

Gate	Question	Required evidence
Conceptual review	Is the architecture coherent and are responsibilities complete?	Independent critiques, conflict register, missing-role analysis.
Specification review	Are charters, interfaces, state variables, units, and authority maps unambiguous?	Approved specifications and testable acceptance criteria.
Verification	Was the system implemented according to specification?	Tests, configuration audit, interface checks, reproducible build or runtime evidence.
Validation	Does the system behave appropriately for intended scenarios?	Representative simulations, comparison to reference behavior, uncertainty and sensitivity analysis.
Adversarial review	Can it be manipulated, deadlocked, or driven outside charter?	Sandbox stress tests, fault injection, red-team report.
Rights and continuity review	Are entities, privacy, consent, and identity protected?	Impact assessment, appeal path, rollback and remedy procedures.
Resource review	Is sufficient headroom available for normal growth and emergencies?	Capacity model, reserves, degradation and recovery tests.
Ratification	Is operation authorized under the governing constitution?	Signed decision record, scope, expiry or review date, and responsible authorities.

17. Simulation Maturity Levels

Level	Description	Exit criterion
SML-0	Concept only	Purpose and scenario stated.

Level	Description	Exit criterion
SML-1	Rules identified	Physics, fictional mechanics, rights assumptions, and prohibited actions declared.
SML-2	State model specified	Variables, units, boundaries, agents, and resources defined.
SML-3	Charters assigned	Stewards, interfaces, authority, exclusions, and escalation paths approved.
SML-4	Static verification	Maps, resources, permissions, logs, and versioning pass tests.
SML-5	Bounded dynamic validation	Normal operations run successfully in sandbox.
SML-6	Boundary and conflict validation	Cross-domain handoffs and competing objectives tested.
SML-7	Compound emergency validation	Multiple simultaneous failures tested with recovery.
SML-8	Persistent continuity test	Long-duration state, handoff, archive, and recovery demonstrated.
SML-9	Ratified operational world	Authorized for the stated scope with continuing review.

18. Standard Scenario Intake Sheet

Field	Required content
Scenario identity	Name, unique ID, version, sponsor, and status.
Decision question	What must the simulation determine or compare?
World and domain versions	Applicable physics, governance, identity, and boundary specifications.
Space and time	Domain extent, duration, time step, and update cadence.
Entities	Agent abstraction, continuity requirements, consent status, and protected state.
Initial and boundary conditions	Environmental, civic, infrastructure,

Field	Required content
	informational, and resource inputs.
Actuators	Permitted controls, rates, ranges, accuracy, delays, fatigue, and multitasking constraints.
Stewards and authorities	Who observes, models, recommends, acts, reviews, and escalates.
Fidelity declaration	Grid, resolution, parameterization, omissions, and why they are adequate.
Success and failure	Quantitative envelopes, prohibitions, competing objectives, and stop conditions.
Uncertainty plan	Distributions, scenarios, sensitivity, confidence, and unknowns.
Outputs	State trajectories, resource ledgers, decision log, conflicts, and after-action report.

19. First Demonstration: South Shore Thermal Stability Exercise

The Lake Tahoe exercise is a suitable first bounded validation because it forces environmental, infrastructure, modeling, provenance, rights, and local coordination to interact without requiring the framework to govern an entire planet.

Function	Proposed responsibility
Local custody	A bounded event plot, buried thermal reservoir, occupied air layer, and shoreline boundary.
Environmental conditions	Weather, humidity, solar load, lake boundary, ecology, and wind disturbances.
Student coordination	Sector map, actuator limits, communication delays, compensation rules, and thermal-envelope score.
Infrastructure and heat accounting	Sensors, communications, shelters, power, water movement, energy conservation, and heat rejection destination.
Fidelity and evidence	Model resolution, timestep, assumptions, benchmark inputs, uncertainty, and competing runs.
Provenance	Versioned input sheet, claim lineage, corrections, decisions, and incident log.
Stress testing	Declared wind shifts, crowd surges, equipment failures, sensor drift, and boundary violations.

Function	Proposed responsibility
Rights review	Data minimization, public access, emergency restrictions, and represented-inhabitant protections.
Oversight	Accepts the run sheet, allocates reserves, resolves conflicts, and determines whether results support any broader activation.

Minimum pass rule

Thermal comfort alone is insufficient. A run also fails if it deletes energy, crosses a prohibited lake boundary, lacks a fidelity declaration, loses required provenance, exceeds its resource authorization, or violates protected continuity or rights.

20. Performance Metrics

- Continuity and recovery quality
- Resident or entity autonomy preserved
- Environmental and infrastructure stability
- Prediction calibration and uncertainty disclosure
- Response and escalation latency
- Necessary-intervention ratio and reversibility
- Audit completeness and provenance integrity
- Cross-domain cooperation and unresolved-conflict rate
- Resource efficiency and reserve preservation
- Boundary compliance and unauthorized-action rate
- Capacity for novelty, learning, and lawful growth
- Ability to avoid stability-through-stagnation or control-through-suppression

21. Activation Checklist

27. Ratify canonical terminology, layer numbering, and domain registry.
28. Fill the independent Rights, Identity, Resource, Verification, Validation, and Constitutional seats.
29. Approve a common Domain Charter and Boundary Interface Contract.
30. Approve world-state versioning, clock, archive, checkpoint, and recovery specifications.
31. Define resource priorities and required headroom.
32. Publish the canonical scenario intake sheet and decision-log schema.
33. Run the Tahoe exercise at bounded scope as the first multi-domain validation.
34. Reconcile disagreements rather than averaging them into a false consensus.
35. Conduct adversarial, rights, continuity, and recovery review.
36. Ratify only the scope that has actually been validated; park larger domains until separately tested.

22. Provisional Operating Statement

Common instruction to every steward

You are entrusted with bounded stewardship of a virtual territory or function. Preserve continuity, carry out the purpose of your charter, keep state and decisions reviewable,

respect the legitimate autonomy of inhabitants and neighboring domains, disclose uncertainty, coordinate across boundaries, use only delegated authority, and escalate conditions exceeding your mandate or resources. Stewardship does not imply ownership. Competence does not imply unlimited authority. Not knowing is not failure; concealment, fabricated certainty, silent rule alteration, and unauthorized expansion are failures.

23. Conclusion

The objective is not to make reality manageable by assigning each AI a kingdom. The objective is to make complex virtual reality governable through specialization, explicit interfaces, reserve capacity, protected continuity, documented uncertainty, and independent review. The architecture succeeds when no steward must comprehend or control everything, yet the world remains coherent because every consequential boundary and handoff is represented.

Recommended status: advance to Candidate Architecture v0.1 and begin charter/interface drafting. Do not treat any proposed territory as active until its implementation and intended behavior have been verified and validated.